

The inductive step

Once the base case is fixed the inductive step carries the entire proof obligation, so no separate base case is needed for each residue class. The step must hold for every n above the base, and that is the whole of it.

Strong induction restated

Strong induction proves a property for every natural number at or above the base case by assuming the property holds for every value strictly below n , rather than only for n minus one. It is a convenience and not a stronger axiom: any proof written with strong induction can be rewritten as ordinary induction over a modified predicate. The base case is not a formality. A proof with a false base case establishes nothing at all, however clean the inductive step looks on the page. Use the strong form when the recursion in the problem reaches back more than a single step, which is the usual situation for divide and conquer recurrences. Ordinary induction hands you one previous case; strong induction hands you all of them.

The well-ordering principle

Every non-empty set of natural numbers has a least element. This principle is logically equivalent to induction and is often taken as the primitive instead, with induction derived from it as a theorem about minimal counterexamples.

Recurrence relations in detail

A recurrence relation expresses a term of a sequence using earlier terms, and solving one means finding a closed form. The inductive step in a divide and conquer proof reaches back more than a single previous case.

Unrelated graph traversal

Breadth first traversal visits every vertex at distance k before any vertex at distance k plus one, using a queue to hold the frontier between the two layers.

A long historical digression

The modern statement of induction is usually credited to Pascal and Fermat, though arguments of the same shape appear far earlier in Euclid and in the work of the Indian mathematician Bhaskara. Formal treatment waited for Dedekind and Peano, who made induction an axiom of arithmetic rather than a technique of proof. The history matters less than the practice, but the vocabulary of the field still carries traces of the older debate about whether induction is discovered or stipulated. Students encountering the axiom for the first time often ask which it is, and the honest answer is that for arithmetic as we use it the question has no operational content.